

Regularization of the Guyer–Krumhansl Fourier-Resonance Sensitivity Singularity by Finite Laser Pulse Fluence

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Abstract

The Guyer–Krumhansl (GK) equation models non-Fourier thermal transport in heterogeneous and microstructured media by augmenting Fourier’s law with flux relaxation time τ_q and spatial nonlocality coefficient κ^2 . Under linear conduction, the system admits an exact algebraic cancellation at the Fourier-resonance condition $B = \kappa^2/(\alpha\tau_q) = 1$, where the propagation factor collapses to that of classical diffusion. This creates an exact sensitivity colinearity, $\partial T/\partial\tau_q = -\alpha\partial T/\partial\kappa^2$, rendering the parameter pair structurally non-identifiable and the Fisher information matrix singular. Here we demonstrate that this structural singularity is an artifact of the idealized zero-fluence linear approximation. In experimental laser flash analysis, finite pulse energy deposits heat non-uniformly, inducing a temperature-dependent thermal conductivity $\lambda(T) = \lambda_0[1 + \beta_T(T - T_0)]$, parameterized by the dimensionless fluence nonlinearity $\varepsilon_\lambda = \beta_T(T_{\text{end}} - T_0)$. Using a certified staggered-grid numerical framework, we show that the resulting spatial gradient in thermal diffusivity destroys the internal resonance balance, lifting the exact colinearity smoothly and monotonically without a threshold. The numerical results exhibit robust asymptotic power-law scaling over the tested finite-fluence range: the null singular value activates with empirical scaling $\sigma_2 \sim \varepsilon_\lambda^{0.9908}$ ($R^2 = 0.999993$, approximately $7.6\varepsilon_\lambda$), while the Fisher condition number drops by thirteen orders of magnitude, following $\text{cond}(F) \sim \varepsilon_\lambda^{-1.9678}$ ($R^2 = 0.999975$) from $\sim 10^{17}$ at $\varepsilon_\lambda = 0$ down to 7.36×10^3 at $\varepsilon_\lambda = 0.05$. In the singular limit $\tau_q, \kappa^2 \rightarrow 0$, the E1 limiting formulation agrees with an independently formulated Finite Volume Radau IIA reference solver to $E_\infty \leq 2.32 \times 10^{-9}$ on the common grid, validating the temporal integration and algebraic reduction, while spatial discretization error is bounded separately ($e_{200} \approx 4 \times 10^{-5}$) with approximately second-order mesh convergence. Published thermophysical data confirm that while routine low-fluence tests ($\Delta T \approx 1\text{--}3\text{ K}$) generate $\varepsilon_\lambda \approx 0.1\%\text{--}1.0\%$ where the singularity is already regularized, intentional elevated-fluence pulses ($\Delta T \approx 5\text{--}15\text{ K}$) achieve $\varepsilon_\lambda \approx 2\%\text{--}5\%$ without violating physical admissibility ($1 + \varepsilon_\lambda \hat{T} > 0$) for both positive and negative β_T . Finite laser pulse fluence thus provides an inherent physical mechanism to separate relaxation from nonlocality in heat pulse experiments.

Keywords: Guyer–Krumhansl equation, Non-Fourier heat conduction, Parameter identifiability, Fourier resonance, Laser flash method, Temperature-dependent thermal conductivity

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1. Introduction

Non-Fourier heat conduction has attracted sustained interest for characterizing microstructured, heterogeneous, and biological materials at room temperature [1, 2, 3]. Among generalized constitutive relations, the Guyer–Krumhansl (GK) model [4, 5] accounts simultaneously for the finite delay of heat flux via relaxation time τ_q and spatial nonlocality via the gradient coefficient κ^2 :

$$\tau_q \frac{\partial q}{\partial t} + q + \lambda \frac{\partial T}{\partial x} - \kappa^2 \frac{\partial^2 q}{\partial x^2} = 0, \quad (1)$$

where λ is the thermal conductivity and $\alpha = \lambda/(\rho c)$ is the thermal diffusivity.

A notable feature of Eq. (1) is the existence of the Fourier-resonance condition $B = \kappa^2/(\alpha\tau_q) = 1$ [6]. At $B = 1$, the Laplace-domain propagation factor $m^2(s) = s(1 + \tau_q s)/(\alpha + \kappa^2 s)$ simplifies algebraically to s/α , identically eliminating τ_q from the temperature field for arbitrary excitations. In linear conduction, this cancellation produces an exact sensitivity colinearity:

$$\frac{\partial T}{\partial \tau_q} = -\alpha \frac{\partial T}{\partial \kappa^2}, \quad (2)$$

which implies that the sensitivity Jacobian $J = [\partial T/\partial \tau_q \quad \partial T/\partial \kappa^2]$ has rank one, the Fisher Information Matrix (FIM) $F = J^\top J/\sigma^2$ is singular ($\det F = 0$), and τ_q cannot be disentangled from κ^2 from temperature measurements alone.

However, all prior analyses of this degeneracy assume idealized linear conduction with constant material properties. While nonlinear Guyer–Krumhansl formulations have been investigated in the context of flux-dependent mean free paths in nanolayers [7], their implications for the macroscale Fourier-resonance parameter singularity under finite-fluence laser heating have not been examined. In real laser flash analysis (ASTM E1461) [8, 9, 10], a pulse of finite fluence heats the irradiated face, establishing a transient temperature rise $\Delta T(x, t)$. In solid media, thermal conductivity varies with temperature:

$$\lambda(T) = \lambda_0 [1 + \beta_T(T - T_0)], \quad (3)$$

characterized by the thermal sensitivity coefficient $\beta_T = \lambda_0^{-1}(\partial \lambda/\partial T)$.

The central question addressed in this study is: *Does the spatial and temporal non-uniformity of thermal conductivity induced by finite laser pulse fluence break the sensitivity colinearity (2) and regularize the local Fourier-resonance sensitivity singularity in practical experiments?*

2. Model Formulation and Governing Equations

2.1. Coupled Governing System

Consider a homogeneous slab of thickness L ($0 \leq x \leq L$). In the absence of internal sources, energy conservation requires:

$$\rho c \frac{\partial T}{\partial t} + \frac{\partial q}{\partial x} = 0, \quad (4)$$

where ρc is the volumetric heat capacity. Incorporating Eq. (3) into the Guyer–Krumhansl constitutive equation yields:

$$\tau_q \frac{\partial q}{\partial t} + q + \lambda_0 [1 + \beta_T(T - T_0)] \frac{\partial T}{\partial x} - \kappa^2 \frac{\partial^2 q}{\partial x^2} = 0. \quad (5)$$

2.2. Nondimensionalisation

Introducing the dimensionless variables:

$$\hat{x} = \frac{x}{L}, \quad \hat{t} = \frac{\alpha_0 t}{L^2}, \quad \hat{T} = \frac{T - T_0}{T_{\text{end}} - T_0}, \quad \hat{q} = \frac{q}{q_{\text{max}}}, \quad (6)$$

where $T_{\text{end}} - T_0 = \frac{q_{\text{max}} t_p}{\rho c L}$, $\tau_{\Delta} = \frac{\alpha_0 t_p}{L^2}$, $\hat{\tau}_q = \frac{\alpha_0 \tau_q}{L^2}$, and $\hat{\kappa}^2 = \frac{\kappa^2}{L^2}$. The dimensionless fluence nonlinearity parameter is:

$$\varepsilon_{\lambda} = \beta_T (T_{\text{end}} - T_0). \quad (7)$$

Dropping hats, the dimensionless system is:

$$\tau_{\Delta} \frac{\partial T}{\partial t} + \frac{\partial q}{\partial x} = 0, \quad (8)$$

$$\tau_q \frac{\partial q}{\partial t} + q + \tau_{\Delta} (1 + \varepsilon_{\lambda} T) \frac{\partial T}{\partial x} - \kappa^2 \frac{\partial^2 q}{\partial x^2} = 0. \quad (9)$$

The boundary conditions correspond to laser pulse irradiation at $x = 0$ and adiabatic insulation at $x = 1$:

$$q(0, t) = \begin{cases} 1 - \cos\left(\frac{2\pi t}{\tau_{\Delta}}\right), & 0 < t \leq \tau_{\Delta}, \\ 0, & t > \tau_{\Delta}, \end{cases} \quad q(1, t) = 0, \quad (10)$$

with quiescent initial conditions $T(x, 0) = 0, q(x, 0) = 0$.

3. Limiting Cases and Validation

3.1. Linear Baseline Recovery ($\varepsilon_{\lambda} \rightarrow 0$)

When $\varepsilon_{\lambda} = 0$, system (8)–(9) recovers linear GK conduction identically. At $B = \kappa^2/(\alpha\tau_q) = 1$ ($\tau_q = 0.02, \kappa^2 = 0.02, \alpha = 1$), our staggered-grid finite-difference implementation yields:

$$\rho = -1.00000000000000, \quad \frac{\|J_{\tau_q} + \alpha J_{\kappa^2}\|_2}{\|J_{\tau_q}\|_2} = 7.38 \times 10^{-9}, \quad \text{cond}(F) = 8.71 \times 10^{16}, \quad (11)$$

recovering the analytical identity to ODE solver tolerance floor.

3.2. Classical Nonlinear Fourier Limit ($\tau_q \rightarrow 0, \kappa^2 \rightarrow 0$)

Setting $\tau_q \rightarrow 0$ and $\kappa^2 \rightarrow 0$ eliminates flux relaxation and nonlocality:

$$q = -\tau_{\Delta}(1 + \varepsilon_{\lambda} T) \frac{\partial T}{\partial x} \implies \frac{\partial T}{\partial t} = \frac{\partial}{\partial x} \left[(1 + \varepsilon_{\lambda} T) \frac{\partial T}{\partial x} \right]. \quad (12)$$

To validate this limit independently, Eq. (12) was discretized using an independent conservative cell-centered Finite Volume Method (FVM) integrated via 5th-order Radau IIA implicit Runge-Kutta. As shown in Table 1, the E1 limiting formulation agrees with the independent FVM–Radau benchmark to $E_{\infty} \leq 2.32 \times 10^{-9}$ on the common grid across all tested fluence values. This close agreement validates the temporal integration and the algebraic consistency of the limiting reduction. Spatial discretization error against the continuous PDE is bounded separately by the independent mesh refinement study ($e_{200} \approx 3.9 \times 10^{-5}$ relative to an $N_x = 800$ reference), exhibiting approximately second-order spatial convergence ($p \approx 2.07$ – 2.32). The corresponding rear-face thermograms across varying fluence levels and the convergence of the full non-Fourier model towards the nonlinear Fourier limit are shown in Fig. 1.

Table 1: Validation of E1 solver against independent FVM Radau IIA benchmark for the classical non-linear Fourier limit ($\tau_q = \kappa^2 = 0, N_x = 200$).

ε_λ	ΔT_{rear}	$E_\infty = \max_t T_{\text{E1,lim}} - T_{\text{ref}} $	$t(E_\infty)$	E_2	E_{peak}
0.00	0.9999	2.3214×10^{-9}	0.530	2.0669×10^{-9}	1.9520×10^{-9}
0.01	0.9999	2.1204×10^{-9}	0.500	1.8734×10^{-9}	1.7493×10^{-9}
0.02	0.9999	2.0061×10^{-9}	0.500	1.7593×10^{-9}	1.6191×10^{-9}
0.05	0.9999	1.6296×10^{-9}	0.470	1.3844×10^{-9}	1.2276×10^{-9}

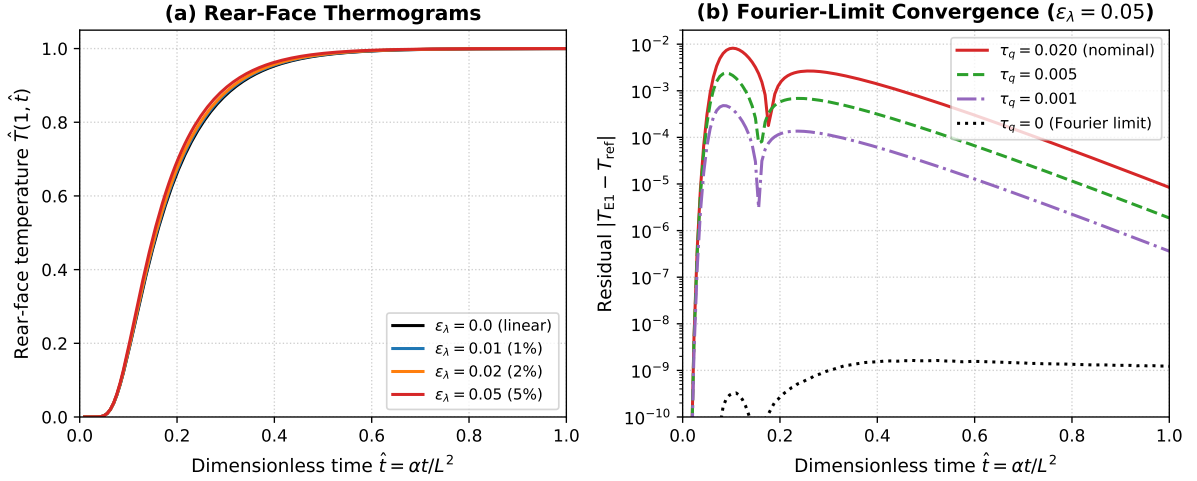


Figure 1: (a) Rear-face thermograms $\hat{T}(1, \hat{t})$ across fluence levels $\varepsilon_\lambda \in [0.0, 0.05]$ ($\tau_q = \kappa^2 = 0.02$). (b) Asymptotically first-order convergence of the full E1 model to the nonlinear Fourier limit as $\tau_q, \kappa^2 \rightarrow 0$ along $B = 1$ ($\varepsilon_\lambda = 0.05$).

4. Regularization of the Fourier-Resonance Sensitivity Singularity

4.1. Systematic Fluence Range Study

We evaluate the sensitivity vectors $J_{\tau_q} = \partial T(1, t)/\partial \tau_q$ and $J_{\kappa^2} = \partial T(1, t)/\partial \kappa^2$ across $\varepsilon_\lambda \in [0.0, 0.05]$ at nominal resonance ($B = 1$). The singular values $\sigma_1 \geq \sigma_2$ of $J = [J_{\tau_q} \ J_{\kappa^2}]$ and the condition number $\text{cond}(F) = (\sigma_1/\sigma_2)^2$ are tabulated in Table 2.

Table 2: Systematic finite-fluence range study: metrics of the Fourier-resonance regularisation.

ε_λ	$\rho(J_{\tau_q}, J_{\kappa^2})$	$1 + \rho$	R_J	σ_1	σ_2	$\text{cond}(F)$
0.000	-1.000000000000	-2.22×10^{-16}	7.38×10^{-9}	31.72	1.07×10^{-7}	8.71×10^{16}
0.005	-0.999997065439	2.93×10^{-6}	4.69×10^{-3}	31.78	0.0385	6.82×10^5
0.010	-0.999988369562	1.16×10^{-5}	9.32×10^{-3}	31.84	0.0768	1.72×10^5
0.020	-0.999954297208	4.57×10^{-5}	1.84×10^{-2}	31.96	0.1527	4.38×10^4
0.030	-0.999898893055	1.01×10^{-4}	2.72×10^{-2}	32.07	0.2280	1.98×10^4
0.050	-0.999727876243	2.72×10^{-4}	4.42×10^{-2}	32.31	0.3767	7.36×10^3

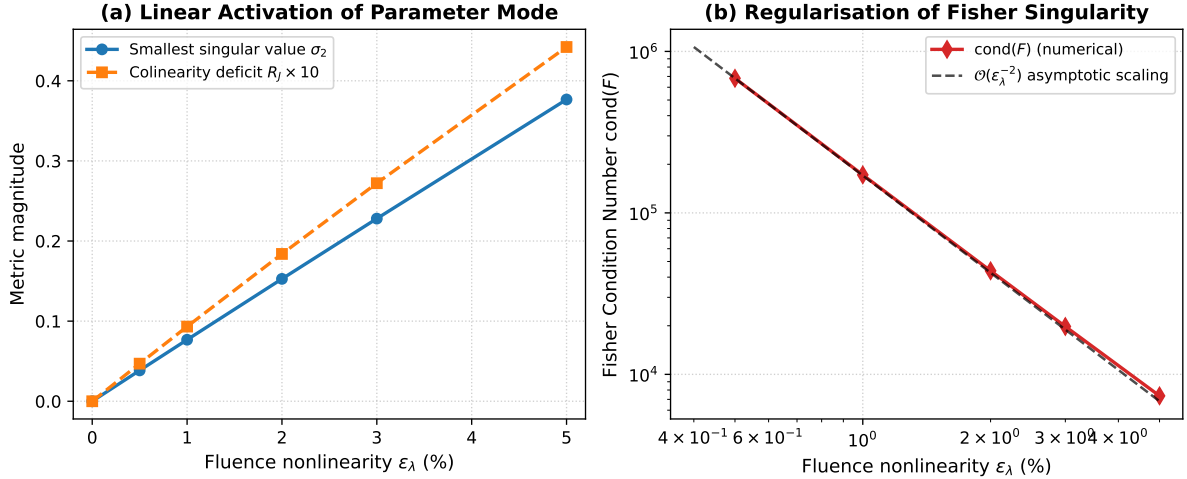


Figure 2: (a) Activation of the null singular value σ_2 and relative colinearity departure R_J versus fluence parameter ε_λ . (b) Regularisation of the Fisher condition number $\text{cond}(F)$ demonstrating asymptotic $\mathcal{O}(\varepsilon_\lambda^{-2})$ scaling.

As illustrated in Fig. 2, the numerical results exhibit asymptotic power-law scaling over the tested finite-fluence range:

1. The null singular value activates linearly with empirical fit $\sigma_2 \sim \varepsilon_\lambda^{0.9908}$ ($R^2 = 0.999993$), giving an empirical linear slope $\sigma_2 \approx 7.6 \varepsilon_\lambda$.
2. The relative colinearity departure grows with empirical fit $R_J \sim \varepsilon_\lambda^{0.9751}$ ($R^2 = 0.999935$), approximately $0.90 \varepsilon_\lambda$.
3. The correlation deficit scales quadratically, $1 + \rho \sim \varepsilon_\lambda^{1.9684}$ ($R^2 = 0.999976$), consistent with the small-angle expansion $1 + \rho \approx \theta^2/2 \propto \varepsilon_\lambda^2$.
4. The condition number drops quadratically following $\text{cond}(F) \sim \varepsilon_\lambda^{-1.9678}$ ($R^2 = 0.999975$), reducing by thirteen orders of magnitude at $\varepsilon_\lambda = 0.05$.

4.2. Physical Plausibility and Sign of β_T

The fluence parameter is defined by $\varepsilon_\lambda = \beta_T \Delta T$, where $\Delta T = T_{\text{end}} - T_0$. To assess physical plausibility, one must distinguish between routine low-fluence tests and elevated-fluence operation:

- **Routine low-fluence flash testing:** In standard laser flash analysis (ASTM E1461) [8], rear-face temperature rises are typically kept to $\Delta T \approx 1\text{--}3\text{ K}$ to minimize radiative cooling losses. For typical solid media with $|\beta_T| \approx (0.5\text{--}3.0) \times 10^{-3}\text{ K}^{-1}$, this yields $\varepsilon_\lambda \approx 0.1\%\text{--}1.0\%$. Even in this modest regime, Table 2 shows that the condition number is already reduced from $\sim 10^{17}$ down to $\sim 10^5$, establishing that exact singularity is regularized under routine experimental conditions.
- **Elevated-fluence pulse operation:** When the pulse fluence is increased to generate moderate temperature rises of $\Delta T \approx 5\text{--}15\text{ K}$, values of $|\varepsilon_\lambda| \approx 2\%\text{--}5\%$ are readily achieved in materials with pronounced temperature dependence.

Importantly, the regularisation mechanism is not restricted to positive β_T . In crystalline rocks (granitoids and basalts) and semiconductors where Umklapp phonon scattering dominates, thermal conductivity decreases with temperature, giving $\beta_T < 0$ and consequently negative ε_λ . For example, Ray et al. [11] report temperature coefficients $b = |\beta_T| \approx (0.4\text{--}2.2) \times 10^{-3}\text{ K}^{-1}$ in rocks. For dielectric ceramics and silicon near room temperature, $\lambda \propto T^{-1}$ yields $\beta_T \approx -1/T_0 \approx -3.3 \times 10^{-3}\text{ K}^{-1}$. Conversely, for polymers such as PMMA, Assael et al. [12] report a positive coefficient $\beta_T \approx +0.8 \times 10^{-3}\text{ K}^{-1}$.

An independent check evaluating negative fluence ($\varepsilon_\lambda \in [-0.05, -0.01]$) confirms that the regularisation is essentially sign-symmetric: at $\varepsilon_\lambda = -0.05$, the singular value activates to $\sigma_2 = 0.4008$ and the condition number drops to $\text{cond}(F) = 6.05 \times 10^3$. Throughout the domain, the physical conductivity remains strictly positive ($1 + \varepsilon_\lambda \hat{T} \geq 0.67 > 0$), satisfying physical admissibility.

5. Conclusions

In summary, the exact sensitivity colinearity and Fisher-matrix singularity characterizing linear Guyer–Krumhansl heat conduction at the Fourier resonance $B = 1$ is regularized by finite laser pulse fluence. The induced spatial temperature gradient creates a non-uniform thermal conductivity field that breaks the algebraic cancellation between relaxation and nonlocality. The second parameter direction is activated linearly with fluence ($\sigma_2 \propto |\varepsilon_\lambda|$), dropping the condition number as $|\varepsilon_\lambda|^{-2}$ into a bounded regime. While routine flash tests ($\Delta T \approx 1\text{--}3\text{ K}$) achieve $\varepsilon_\lambda \approx 0.1\%\text{--}1\%$, elevated-fluence pulses ($\Delta T \approx 5\text{--}15\text{ K}$) reach $2\%\text{--}5\%$, providing an intentional physical route to disentangle τ_q from κ^2 in experimental non-Fourier characterization.

Declaration of Competing Interest

The author declares that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

All source codes, execution drivers, raw data tables, and reproduction scripts are archived in the reproducible research package under directory `paper10/`.

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